Project Part 1

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Repository URL: https://github.com/UVA-STAT3080/student-Sp26

Introduction

The unemployment rate is one of the most closely watched indicators of economic health in the United States. While national figures dominate headlines, state-level rates can vary substantially, reflecting differences in industry composition, demographics, and regional economic conditions. This report investigates the following research question: **How do annual unemployment rates vary across U.S. states and Census regions, and how have these rates changed from 2019 to 2024?**

This time window spans the pre-pandemic economy (2019), the COVID-19 labor market shock (2020), and the subsequent recovery period (2021–2024). Understanding geographic patterns in unemployment can inform economic policy and highlight which regions are most vulnerable to major labor market disruptions.

Data Summary

The data come from the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) program, a federal-state cooperative effort administered by the U.S. Department of Labor. The program produces annual estimates of unemployment for all 50 U.S. states and the District of Columbia, derived by combining data from the Current Population Survey (CPS), the Current Employment Statistics (CES) survey, and state unemployment insurance (UI) systems. Because estimates are produced for every state and D.C., the data represent the entire target population of U.S. state-level jurisdictions, not a sample.

Annual average unemployment rates for 2019 through 2024 were compiled from four official BLS annual averages news releases (see References). A Region variable was added using the U.S. Census Bureau's standard four-region classification (Northeast, South, Midwest, West).

Two limitations are worth noting. First, LAUS estimates are periodically revised as benchmark data become available. Second, the unemployment rate excludes discouraged workers who have stopped searching for employment, which may understate true labor market weakness during severe downturns. Despite these limitations, the data are appropriate for the research question because they provide consistent, official government estimates across three economically distinct periods.

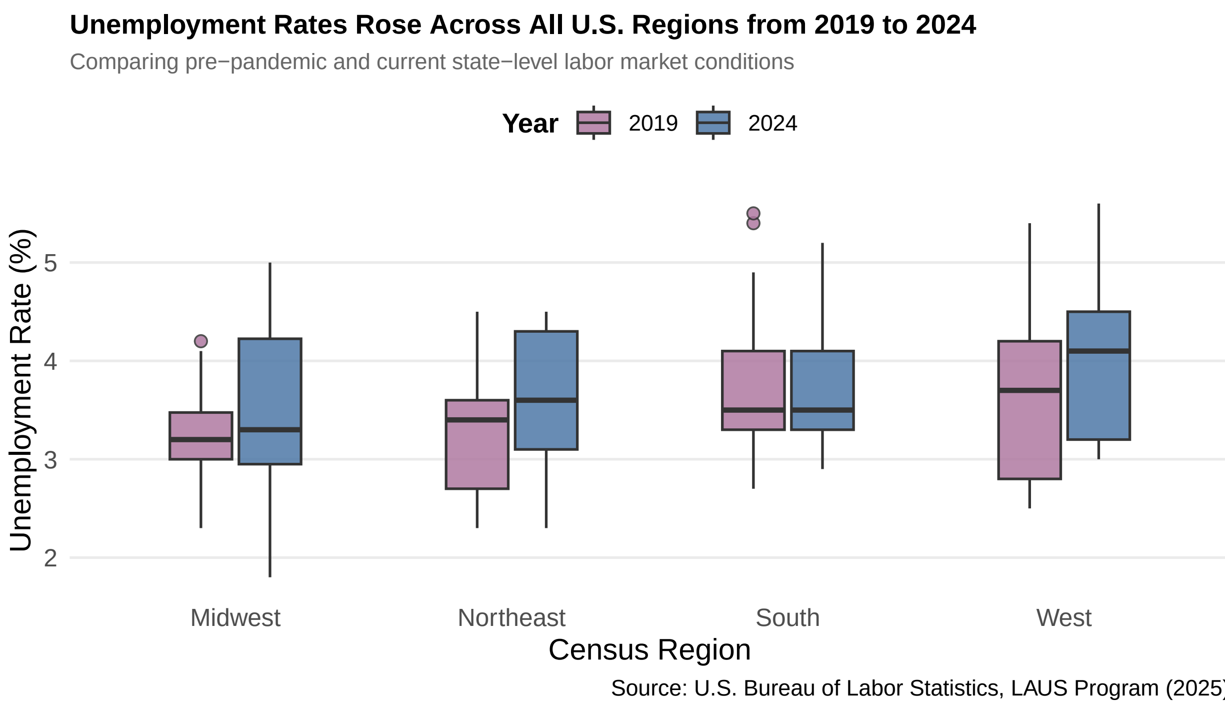
Exploratory Analysis

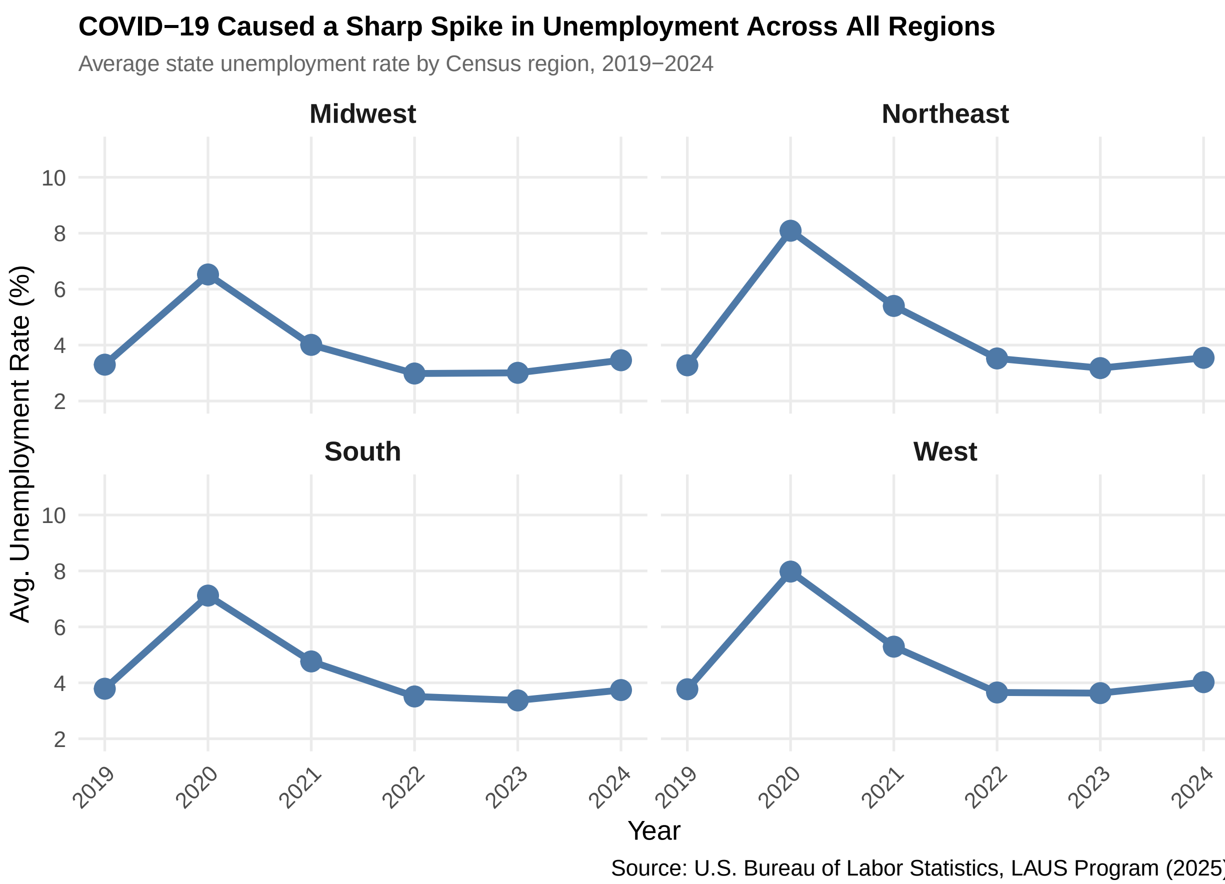
Table 1: **Summary Statistics of State Unemployment Rates (%) by Census Region, 2024**

Region	N	Mean	Median	Std Dev	Min	Max
West	13	4.02	4.1	0.84	3.0	5.6
South	17	3.74	3.5	0.68	2.9	5.2
Northeast	9	3.54	3.6	0.79	2.3	4.5
Midwest	12	3.46	3.3	0.96	1.8	5.0

Table 2: **Five Highest and Five Lowest State Unemployment Rates, 2024**

State	Census Region	Rate (%)	Group
Nevada	West	5.6	Highest 5
California	West	5.3	Highest 5
District of Columbia	South	5.2	Highest 5
Kentucky	South	5.1	Highest 5
Illinois	Midwest	5.0	Highest 5
South Dakota	Midwest	1.8	Lowest 5
Vermont	Northeast	2.3	Lowest 5
North Dakota	Midwest	2.4	Lowest 5
New Hampshire	Northeast	2.6	Lowest 5
Nebraska	Midwest	2.8	Lowest 5


Unemployment Rates Rose Across All U.S. Regions from 2019 to 2024
Comparing pre−pandemic and current state−level labor market conditions
Year
2019
2024
Unemployment Rate (%)
5
4
3
2
Midwest
Northeast
South
West
Census Region
Source: U.S. Bureau of Labor Statistics, LAUS Program (2025)


COVID−19 Caused a Sharp Spike in Unemployment Across All Regions
Average state unemployment rate by Census region, 2019−2024
Midwest
Northeast
South
West
Avg. Unemployment Rate (%)
10
8
6
4
2
2019
2020
2021
2022
2023
2024
Year
Source: U.S. Bureau of Labor Statistics, LAUS Program (2025)

Conclusions

The exploratory analysis reveals several notable patterns in U.S. state-level unemployment rates across the 2019–2024 period.

Regional variation. Table 1 shows that the West had the highest mean unemployment rate in 2024 at 4.02%, driven largely by California and Nevada, whose large labor forces are concentrated in industries sensitive to economic cycles. The Midwest had the lowest mean rate at 3.46% and the smallest standard deviation, indicating both lower unemployment and greater consistency across its states.

State-level extremes. Table 2 confirms that the five highest-unemployment states in 2024 were Nevada (5.6%), California (5.3%), District of Columbia (5.2%), Kentucky (5.1%), and Illinois (5.0%). The five lowest were South Dakota (1.8%), Vermont (2.3%), North Dakota (2.4%), New Hampshire (2.6%), and Nebraska (2.8%), predominantly small Midwest and Northeast states with tight labor markets.

COVID-19 shock and recovery. Figure 2 shows a dramatic spike in 2020 across all four regions. The West experienced the sharpest increase, driven by Hawaii and Nevada whose tourism-dependent economies were hit hardest by pandemic shutdowns. All regions recovered close to pre-pandemic levels by 2022, though the West remained the highest through 2024.

Implications for Part 2. These patterns suggest that Census region explains a meaningful portion of state-level unemployment variation. A formal test of whether mean unemployment rates differ significantly across regions will be pursued in Part 2.

References

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AI Attribution: Portions of this report including the R Markdown structure, code, and written prose were drafted with assistance from Claude (Anthropic, claude.ai). All data values were verified against official BLS publications. All output was reviewed and edited by the author in accordance with UVA academic integrity guidelines.